

1202SCG Calculus 1
1020ENG Engineering Mathematics 1

Week 8 Lecture

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Table of Integrals

In the table below, n and k are constants, with $n \neq -1$.

Function $f(x)$	Integral $\int f(x) dx$
k	$kx + C$
x^n	$\frac{x^{n+1}}{n+1} + C$
$\sin(x)$	$-\cos(x) + C$
$\cos(x)$	$\sin(x) + C$
e^x	$e^x + C$
$\frac{1}{x}$	$\ln x + C$

Expanded Table of Integrals

Function $f(x)$	Integral $\int f(x) dx$
$\sec^2(x)$	$\tan(x) + C$
$\sinh(x)$	$\cosh(x) + C$
$\cosh(x)$	$\sinh(x) + C$
$\operatorname{sech}^2(x)$	$\tanh(x) + C$

A Quick Check

Integration reverses differentiation.

$$\frac{d}{dx} \left(\int f(x) dx \right) = f(x).$$

For example,

$$\int (3x^2 - 4 \cos x + 2e^x) dx = x^3 - 4 \sin x + 2e^x + C.$$

Differentiate the answer to check it.

Integration by Substitution

Substitution reverses the chain rule.

$$\int f(u(x)) u'(x) dx = \int f(u) du.$$

A substitution succeeds when it replaces the entire integral, including dx , with an expression in the new variable.

$$du = u'(x) dx.$$

Substitution: Example 1

Evaluate

$$\int e^{4x-3} dx.$$

Let $u = 4x - 3$. Then $du = 4 dx$, so $dx = \frac{1}{4} du$.

$$\begin{aligned}\int e^{4x-3} dx &= \frac{1}{4} \int e^u du \\ &= \frac{1}{4} e^u + C \\ &= \frac{1}{4} e^{4x-3} + C.\end{aligned}$$

Substitution: Example 2

Evaluate

$$\int x \sin(x^2) dx.$$

Let $u = x^2$. Then $du = 2x dx$, so $dx = \frac{du}{2x}$.

$$\begin{aligned}\int x \sin(x^2) dx &= \int x \sin u \left(\frac{du}{2x} \right) \\ &= \frac{1}{2} \int \sin u du \\ &= -\frac{1}{2} \cos(x^2) + C.\end{aligned}$$

Substitution: Example 3

Evaluate

$$\int_0^2 (3x + 1)^2 dx.$$

Let $u = 3x + 1$. Then $du = 3 dx$, so $dx = \frac{1}{3}du$. The limits become

$$x = 0 \Rightarrow u = 1, \quad x = 2 \Rightarrow u = 7.$$

Therefore,

$$\int_0^2 (3x + 1)^2 dx = \frac{1}{3} \int_1^7 u^2 du = \left[\frac{u^3}{9} \right]_1^7 = 38.$$

Substitution: Example 4

Evaluate

$$\int_0^2 \frac{dx}{5-2x}.$$

Let $u = 5 - 2x$. Then $du = -2 dx$, so $dx = -\frac{1}{2} du$. The limits become

$$x = 0 \Rightarrow u = 5, \quad x = 2 \Rightarrow u = 1.$$

Therefore,

$$\begin{aligned} \int_0^2 \frac{dx}{5-2x} &= -\frac{1}{2} \int_5^1 \frac{1}{u} du \\ &= -\frac{1}{2} [\ln |u|]_5^1 = \frac{1}{2} \ln 5. \end{aligned}$$

Trigonometric Substitution

Trigonometric identities can remove square roots involving sums or differences of squares.

Expression	Substitution	Identity
$\sqrt{a^2 - x^2}$	$x = a \sin \theta$	$1 - \sin^2 \theta = \cos^2 \theta$
$\sqrt{a^2 + x^2}$	$x = a \tan \theta$	$1 + \tan^2 \theta = \sec^2 \theta$

Sine Substitution: Example 1

Evaluate

$$\int \frac{dx}{\sqrt{9-x^2}}.$$

Let $x = 3 \sin \theta$. Then $dx = 3 \cos \theta d\theta$ and $\sqrt{9-x^2} = 3 \cos \theta$.

$$\begin{aligned}\int \frac{dx}{\sqrt{9-x^2}} &= \int \frac{3 \cos \theta}{3 \cos \theta} d\theta \\ &= \theta + C \\ &= \sin^{-1} \left(\frac{x}{3} \right) + C.\end{aligned}$$

Sine Substitution: Example 2

Evaluate $\int \frac{x^2}{\sqrt{16-x^2}} dx$ using $x = 4 \sin \theta$.

$$dx = 4 \cos \theta d\theta,$$

$$\sqrt{16-x^2} = 4 \cos \theta,$$

$$\begin{aligned}\int \frac{x^2}{\sqrt{16-x^2}} dx &= 16 \int \sin^2 \theta d\theta \\ &= 8 \int (1 - \cos 2\theta) d\theta \\ &= 8\theta - 4 \sin 2\theta + C.\end{aligned}$$

Since $\theta = \sin^{-1}(x/4)$ and $\sin 2\theta = 2 \sin \theta \cos \theta = \frac{x\sqrt{16-x^2}}{8}$,

$$\int \frac{x^2}{\sqrt{16-x^2}} dx = 8 \sin^{-1} \left(\frac{x}{4} \right) - \frac{x}{2} \sqrt{16-x^2} + C.$$

Sine Substitution: Example 3

Evaluate

$$\int \frac{dx}{\sqrt{25 - 4x^2}}.$$

Let $x = \frac{5}{2} \sin \theta$. Then

$$dx = \frac{5}{2} \cos \theta \, d\theta,$$

$$\sqrt{25 - 4x^2} = 5 \cos \theta.$$

Therefore,

$$\int \frac{dx}{\sqrt{25 - 4x^2}} = \frac{1}{2} \int d\theta = \frac{1}{2} \sin^{-1} \left(\frac{2x}{5} \right) + C.$$

Tangent Substitution: Example 1

Evaluate

$$\int \frac{dx}{x^2 + 4}.$$

Let $x = 2 \tan \theta$. Then $dx = 2 \sec^2 \theta d\theta$ and $x^2 + 4 = 4 \sec^2 \theta$.

$$\begin{aligned} \int \frac{dx}{x^2 + 4} &= \int \frac{2 \sec^2 \theta}{4 \sec^2 \theta} d\theta \\ &= \frac{1}{2} \theta + C \\ &= \frac{1}{2} \tan^{-1} \left(\frac{x}{2} \right) + C. \end{aligned}$$

Tangent Substitution: Example 2

Evaluate

$$\int \frac{dx}{2x^2 + 50}.$$

Let $x = 5 \tan \theta$. Then $dx = 5 \sec^2 \theta d\theta$ and $2x^2 + 50 = 50 \sec^2 \theta$.

$$\begin{aligned} \int \frac{dx}{2x^2 + 50} &= \frac{1}{10} \int d\theta \\ &= \frac{1}{10} \theta + C \\ &= \frac{1}{10} \tan^{-1} \left(\frac{x}{5} \right) + C. \end{aligned}$$

Completing the Square

A quadratic denominator can often be rewritten in one of the forms

$$(x - h)^2 + a^2 \quad \text{or} \quad a^2 - (x - h)^2.$$

After completing the square, use a tangent substitution for $(x - h)^2 + a^2$ or a sine substitution for $a^2 - (x - h)^2$.

Completing the Square: Example

Evaluate

$$\int \frac{dx}{x^2 + 6x + 13}.$$

Complete the square:

$$x^2 + 6x + 13 = (x + 3)^2 + 4.$$

Let $u = x + 3$. Then

$$\begin{aligned} \int \frac{dx}{(x + 3)^2 + 2^2} &= \frac{1}{2} \tan^{-1} \left(\frac{u}{2} \right) + C \\ &= \frac{1}{2} \tan^{-1} \left(\frac{x + 3}{2} \right) + C. \end{aligned}$$

Completing the Square: Sine Form

Evaluate

$$\int \frac{dx}{\sqrt{5 + 4x - x^2}}.$$

Complete the square:

$$5 + 4x - x^2 = 9 - (x - 2)^2.$$

Let $u = x - 2$. Then

$$\begin{aligned} \int \frac{du}{\sqrt{3^2 - u^2}} &= \sin^{-1} \left(\frac{u}{3} \right) + C \\ &= \sin^{-1} \left(\frac{x - 2}{3} \right) + C. \end{aligned}$$

Integration by Parts

Integration by parts reverses the product rule.

$$\frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}.$$

Rearranging gives

$$u \frac{dv}{dx} = \frac{d}{dx}(uv) - v \frac{du}{dx}.$$

Integrating both sides gives

$$\int u \, dv = uv - \int v \, du.$$

Choose u so that differentiating it makes the remaining integral simpler.

Integration of Parts: Example 1

Evaluate

$$\int xe^x dx.$$

Choose $u = x$ and $dv = e^x dx$. Then $du = dx$ and $v = e^x$.

$$\begin{aligned}\int xe^x dx &= xe^x - \int e^x dx \\ &= xe^x - e^x + C \\ &= e^x(x - 1) + C.\end{aligned}$$

Integration of Parts: Example 2

Evaluate

$$\int \ln x \, dx.$$

Regard the integrand as $(\ln x)(1)$. Choose

$$u = \ln x, \quad dv = dx, \quad du = \frac{1}{x} dx, \quad v = x.$$

Therefore,

$$\begin{aligned} \int \ln x \, dx &= x \ln x - \int x \left(\frac{1}{x} \right) dx \\ &= x \ln x - x + C. \end{aligned}$$

Integration of Parts: Example 3

Evaluate $\int x^2 e^x dx$.

$$\begin{aligned}\int x^2 e^x dx &= x^2 e^x - \int 2x e^x dx \\&= x^2 e^x - 2 \left(x e^x - \int e^x dx \right) \\&= x^2 e^x - 2x e^x + 2e^x + C \\&= e^x (x^2 - 2x + 2) + C.\end{aligned}$$

Each application lowers the degree of the polynomial.

Integration of Parts: Example 4

Evaluate

$$\int_0^{\pi/2} x \cos x \, dx.$$

Choose $u = x$ and $dv = \cos x \, dx$, so $du = dx$ and $v = \sin x$.

$$\begin{aligned} \int_0^{\pi/2} x \cos x \, dx &= [x \sin x]_0^{\pi/2} - \int_0^{\pi/2} \sin x \, dx \\ &= \frac{\pi}{2} - 1. \end{aligned}$$

Integration of Parts: Example 5

Let $I = \int e^x \cos x \, dx$. Applying integration by parts twice,

$$\begin{aligned} I &= e^x \cos x + \int e^x \sin x \, dx \\ &= e^x \cos x + e^x \sin x - I. \end{aligned}$$

The original integral returns. Solve algebraically:

$$\begin{aligned} 2I &= e^x(\sin x + \cos x), \\ I &= \frac{e^x}{2}(\sin x + \cos x) + C. \end{aligned}$$

The constant of integration is added after solving for I .

Choosing an Integration Method

Pattern	Likely method
$f(g(x))g'(x)$	Substitute $u = g(x)$
$\sqrt{a^2 - x^2}$	Use $x = a \sin \theta$
$x^2 + a^2$	Use $x = a \tan \theta$
Quadratic denominator	Complete the square, then substitute
Product such as $x^n e^x$, $x^n \sin x$, or $\ln x$	Use integration by parts

More than one method may work. Prefer the method that simplifies the integral.

Mixed Practice

Identify a suitable method, then evaluate.

1. $\int \frac{x}{(x^2 + 1)^2} dx$

2. $\int \frac{dx}{x^2 + 2x + 5}$

3. $\int \frac{dx}{\sqrt{9 - x^2}}$

4. $\int x \cos(2x) dx$

Mixed Practice: Answers

1. Substitution: $-\frac{1}{2(x^2 + 1)} + C$

2. Complete the square: $\frac{1}{2} \tan^{-1} \left(\frac{x+1}{2} \right) + C$

3. Sine substitution: $\sin^{-1} \left(\frac{x}{3} \right) + C$

4. Integration by parts: $\frac{x}{2} \sin(2x) + \frac{1}{4} \cos(2x) + C$

Week 8 Summary

- ▶ Substitution reverses the chain rule.
- ▶ Sine and tangent substitutions simplify common square-root forms.
- ▶ Completing the square exposes a standard trigonometric form.
- ▶ Integration by parts reverses the product rule.
- ▶ Differentiating an antiderivative provides a reliable check.